

A trained network that could do better

1. Tune the hyperparameters

Number of hidden layers

Neurons per layer

Learning rate

Optimizer

Batch size

Activation function

Epochs

2. Fix the problems

problem

Vanishing or
exploding gradients

fixed by

weight initialisation, activation functions,
batch normalisation, gradient clipping

Not enough data

transfer learning,
unsupervised pre-training

Slow training

optimizers,
learning rate scheduler

Overfitting

L1 and L2 regularisation, dropout,
early stopping